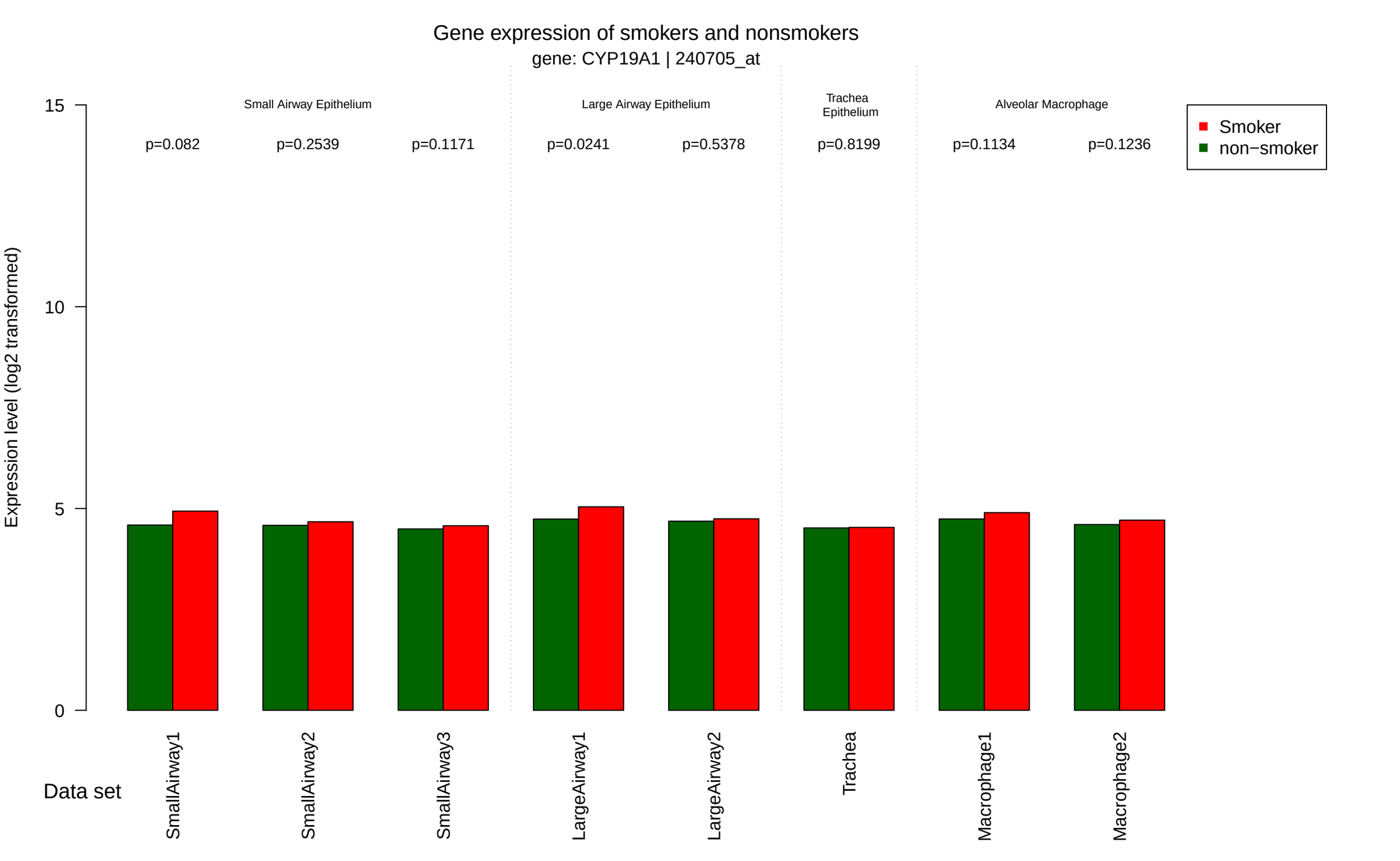
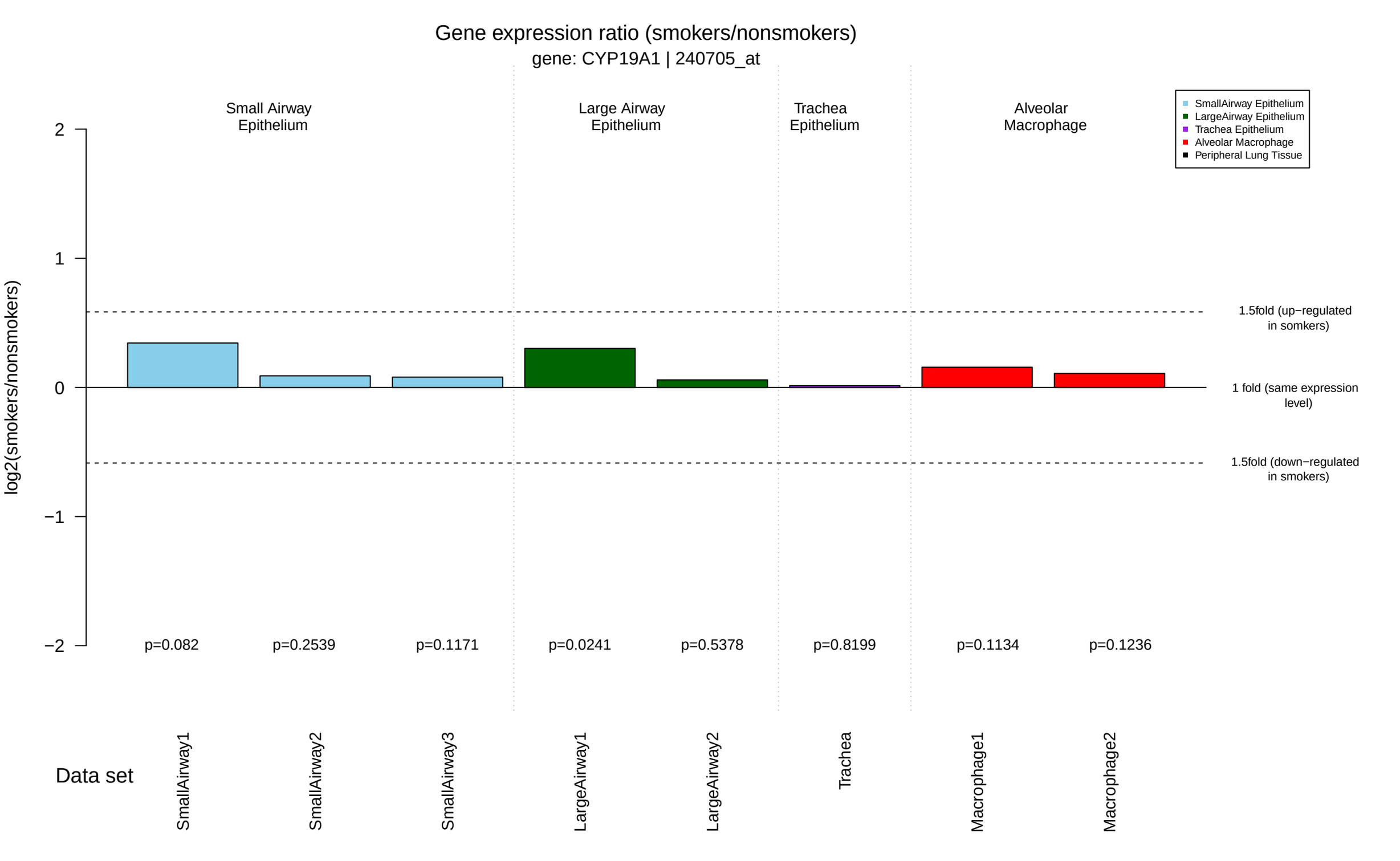
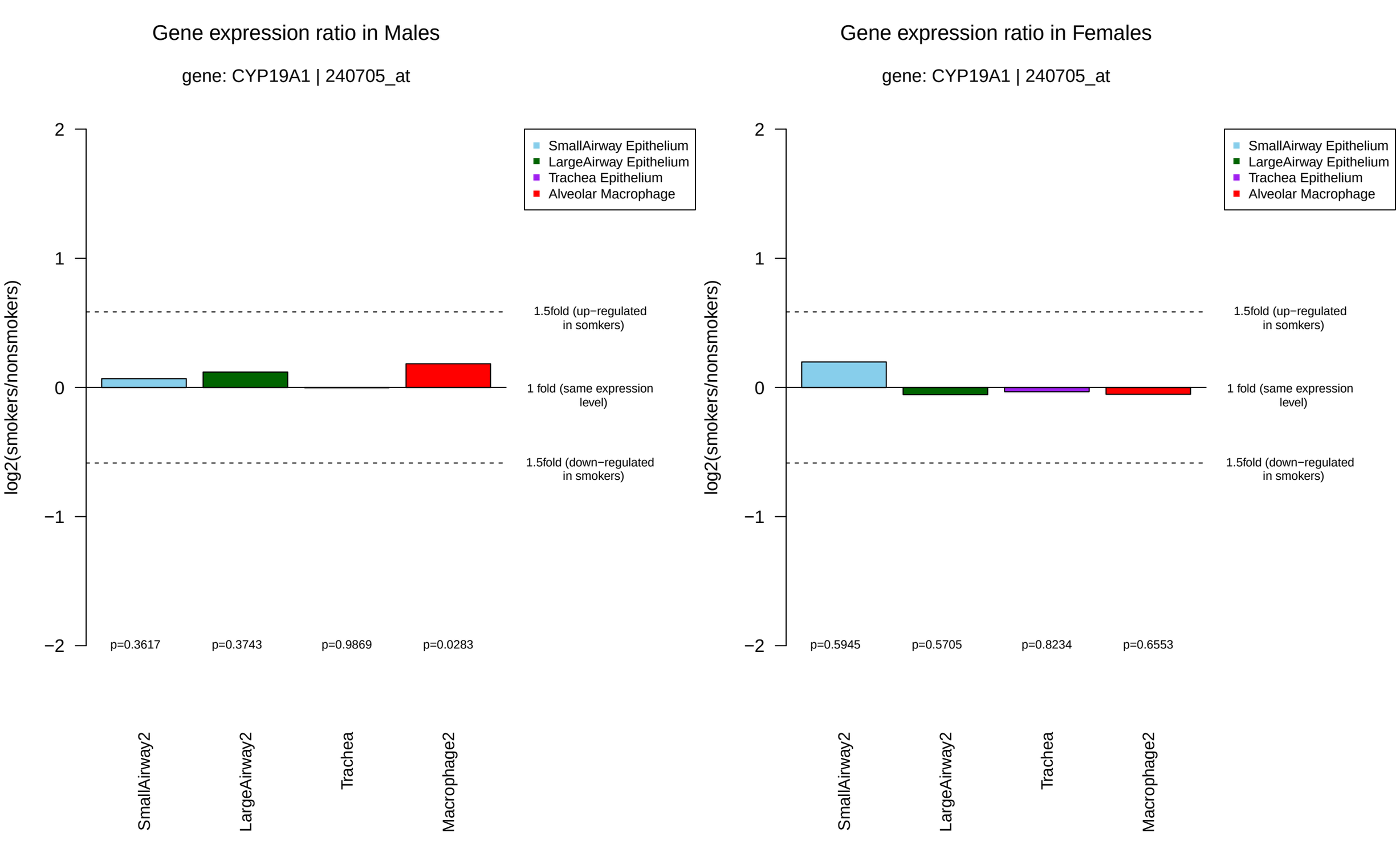
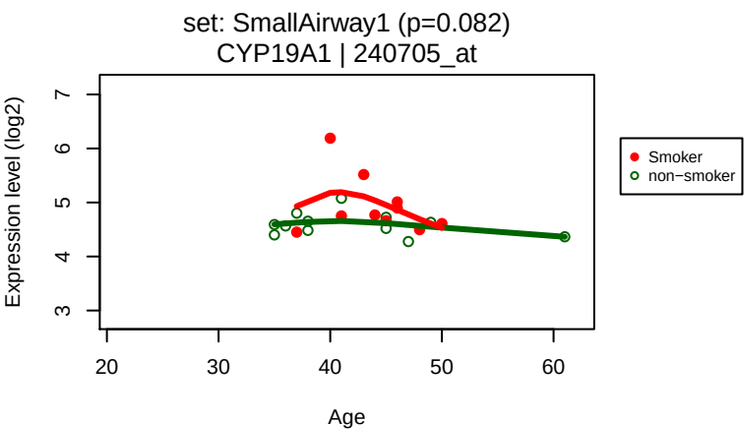
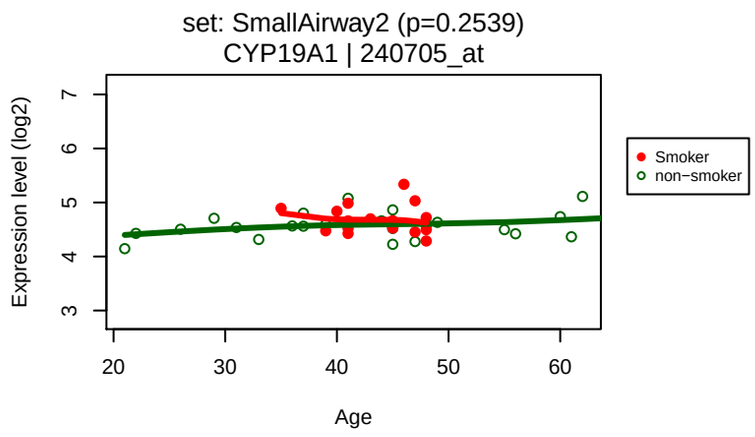
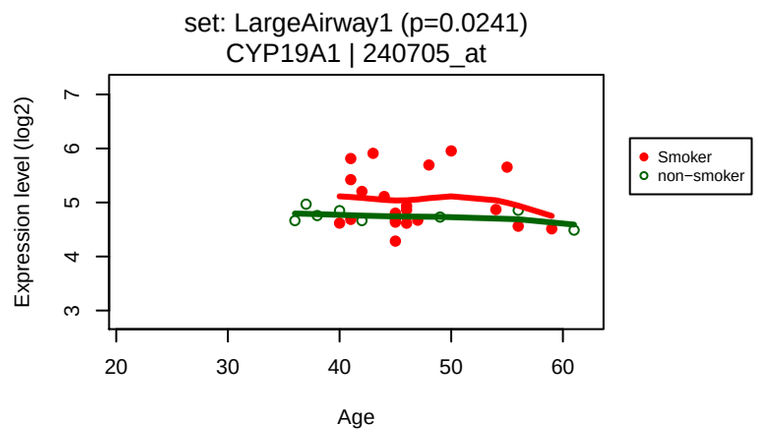

Gene expression of smokers and nonsmokers
gene: CYP19A1 | 240705_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.082
p=0.2539
p=0.1171
p=0.0241
p=0.5378
p=0.8199
p=0.1134
p=0.1236
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

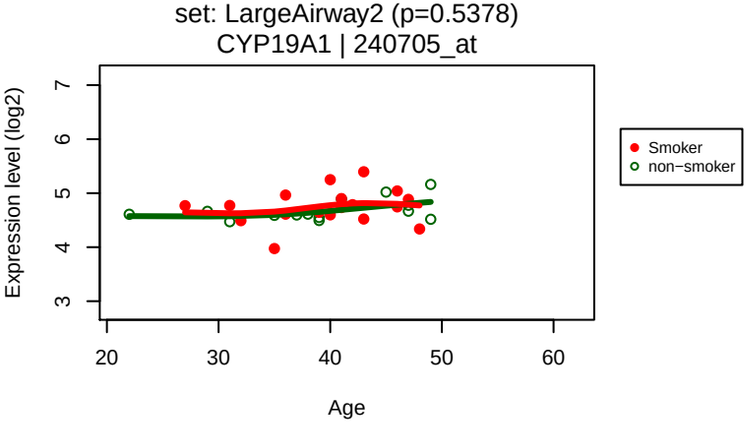

Gene expression ratio (smokers/nonsmokers)
gene: CYP19A1 | 240705_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.082
p=0.2539
p=0.1171
p=0.0241
p=0.5378
p=0.8199
p=0.1134
p=0.1236
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

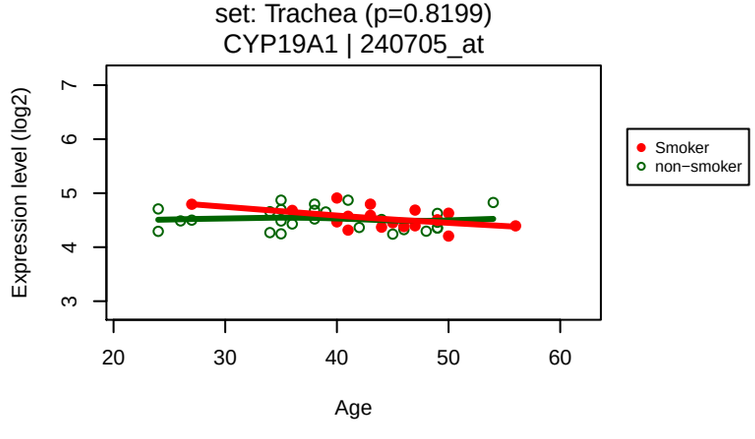

Gene expression ratio in Males
gene: CYP19A1 | 240705_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3617
p=0.3743
p=0.9869
p=0.0283
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP19A1 | 240705_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5945
p=0.5705
p=0.8234
p=0.6553
SmallAirway2
LargeAirway2
Trachea
Macrophage2

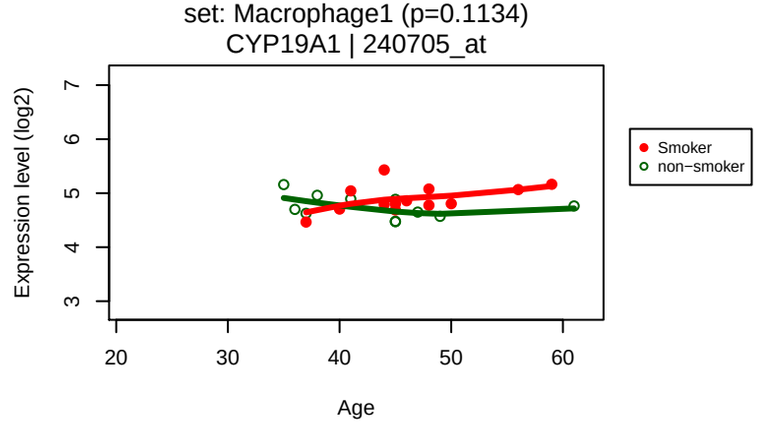

set: SmallAirway1 (p=0.082)
CYP19A1 | 240705_at
Expression level (log2)
Age
Smoker
non−smoker

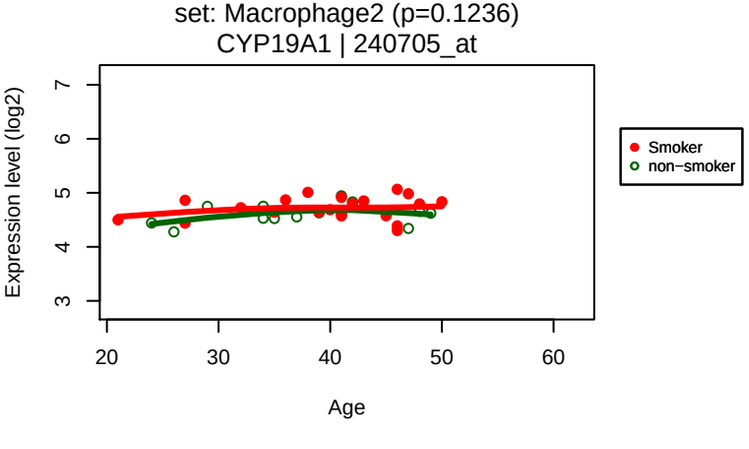

set: SmallAirway2 (p=0.2539)
CYP19A1 | 240705_at
Expression level (log2)
Age
Smoker
non−smoker

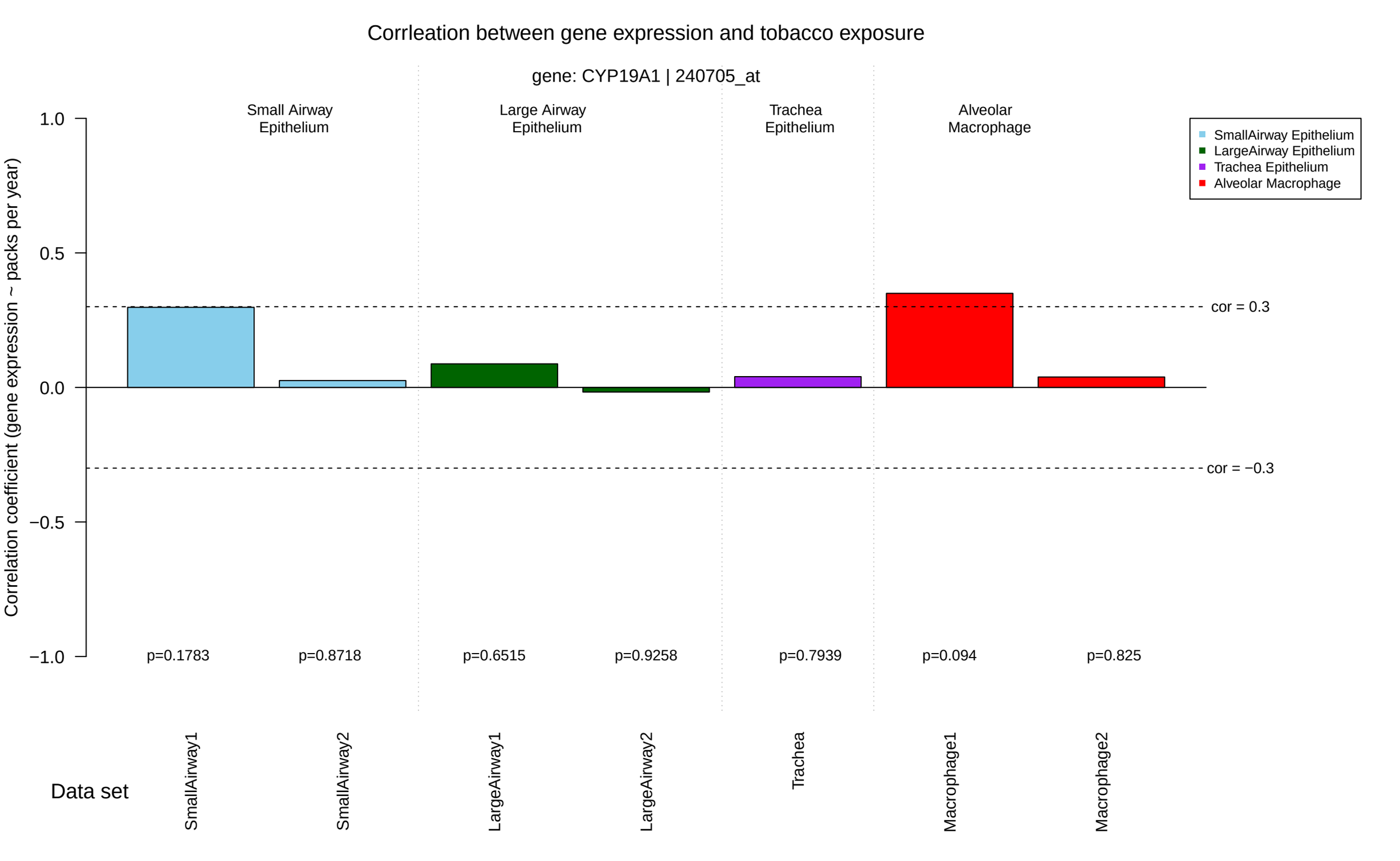

set: LargeAirway1 (p=0.0241)
CYP19A1 | 240705_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5378)
CYP19A1 | 240705_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8199)
CYP19A1 | 240705_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1134)
CYP19A1 | 240705_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1236)
CYP19A1 | 240705_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP19A1 | 240705_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1783
p=0.8718
p=0.6515
p=0.9258
p=0.7939
p=0.094
p=0.825
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2